ZOVAK · PRODUCT PLAN

Aisle — the store that shops with you

What can be automated in online shopping (for shoppers and for store owners), how the agents work, what always needs a human tap, and how we build the backend on what we learned from Nofa / ZOVAK.

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Prepared for Amartya Ghosh · Zovak

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01 Summary

Aisle is an online store with an AI agent at the counter. Shoppers say what they need in their own words; the agent finds three good picks, compares them, recommends a size, fills the cart and handles tracking, swaps and returns. Store owners get a copilot that writes product pages, recovers abandoned carts, forecasts stock and turns reviews into fixes.

The rule from Nofa carries over: **the agent prepares, the person confirms.** Payments, refunds, address changes, price changes and purchase orders always need a human tap.

Aisle can be sold two ways: (a) as our own demo store, and (b) as an agent layer that plugs into an existing Shopify / WooCommerce store. Section 13 asks you to choose.

> Status: the website (aisle-shopping) is built with a working chat + voice widget in demo mode (sample catalogue). The widget calls POST /api/agent/chat, which returns 501 until the backend exists.

02 Who uses it

Person	Pain today	What Aisle does
Shopper who knows the need, not the product	Nine filters, 300 results, no idea which fits	One sentence → three picks with reasons
Shopper unsure about size	Orders two sizes, returns one	Size recommendation with confidence
Gift buyer	Doesn't know the recipient's taste	Budget-bound gift ideas, wrapping, delivery date
Existing customer	Waits for support on tracking/returns	Instant tracking, swaps, refunds in chat, voice, WhatsApp
Store owner	Writing listings, chasing carts, guessing stock	Store copilot: listings, recovery, forecasts, insights
Support team	Repetitive "where is my order" tickets	Agent handles routine; hard cases arrive with context

03 What can be automated

Autonomy levels: **Auto** = agent acts alone (read-only or low-risk); **Propose** = agent prepares, a person taps approve; **Human** = agent only assists.

For the shopper

Task	What the agent does	Agent	Autonomy
Understand the need	Turns a sentence (or voice note, or photo) into structured intent: use, budget, size, deadline, style	Personal shopper	Auto
Search & curate	Searches catalogue + stock + delivery dates; returns 3 picks with a reason each	Curator	Auto
Visual search	"Find something like this" from a photo or Instagram link	Curator	Auto
Compare	Side-by-side on what the shopper said matters, with real review quotes	Compare	Auto
Size & fit	Uses brand size maps, past orders/returns and review signals; gives size + confidence	Size & fit	Auto
Bundles & styling	Completes the look; suggests gifts within budget	Stylist	Auto
Price-drop / restock alerts	Watches items and messages the shopper	Deal watcher	Propose (opt-in)
Cart building	Adds items, applies best valid coupon, picks delivery slot	Checkout	Auto
Payment	Prepares the payment summary; payment happens on the store's checkout	Checkout	Human (tap to pay)
Order tracking	Live status from the courier, delivery window, courier contact	Order care	Auto
Swaps & returns	Checks policy window, books pickup, issues label	Order care	Propose
Refunds	Prepares refund per policy	Order care	Propose (shopper) + rules / staff above a limit
Reorder	"Order my usual" for consumables	Personal shopper	Propose

For the store owner

Task	What the agent does	Agent	Autonomy
Catalogue enrichment	Writes titles, descriptions, tags, size charts, alt text from photos + spec sheets	Store copilot	Propose
Translation	Bangla / Hindi / English listings	Store copilot	Propose
Abandoned-cart recovery	Messages the shopper, answers the real question (size? delivery?), offers help	Store copilot	Auto (within opt-in + frequency caps)
Stock forecasting	Predicts sell-outs by size/colour; drafts purchase orders	Store copilot	Propose

Task	What the agent does	Agent	Autonomy
Review insights	Clusters reviews into themes ("runs narrow") with counts and trends	Store copilot	Auto
Pricing & offers	Suggests bundles/discounts inside margin rules	Store copilot	Propose
Support deflection	Answers order status, policy, sizing; escalates with summary	Order care	Auto
Fraud / abuse flags	Flags risky orders and return abuse for review	Store copilot	Human
Weekly report	Sales, conversion, top questions, lost-sale reasons	Store copilot	Auto

04 The agents

The shopper talks to one assistant (the orchestrator). It hands work to specialists that each own a few tools. All of them share one Shopper profile and one Cart — so the size agent knows what the curator picked, and order care knows what was bought.

Agent	Owns	Main tools (functions)
Orchestrator	Conversation, identity, routing, hand-offs	identify_shopper, route_to_agent, escalate_to_human
Personal shopper	Intent and taste memory	parse_intent, get_profile, remember_preference (consented)
Curator	Search and picks	search_catalog(filters, query), get_stock, get_delivery_estimate, rank_picks
Compare	Side-by-side	get_product, summarise_reviews(product, aspects)
Size & fit	Size recommendation	get_size_chart, get_order_history, review_fit_signal, recommend_size
Stylist	Bundles and gifts	complete_the_look, gift_ideas(budget, recipient)
Deal watcher	Alerts	create_price_alert (gated), check_price_history
Checkout	Cart and payment hand-off	add_to_cart, apply_best_coupon, choose_delivery_slot, create_checkout_link (payment by shopper)
Order care	After-sale	get_order, track_shipment, check_return_policy, propose_swap/return/refund (gated)
Store copilot	Merchant automation	draft_listing, forecast_stock, propose_purchase_order, review_insights, recover_cart

How a purchase flows (example)

```
shopper: "running shoes for flat feet, under $120, before Saturday"
  -> Personal shopper: intent = {use: running, support: stability, budget<=120, deadline: Sat,
size: UK9 (history)}
  -> Curator: search_catalog + get_stock + get_delivery_estimate -> rank_picks -> 3 product
cards
  -> shopper: "what size?" -> Size & fit: UK 9 wide, confidence 86% (reason shown)
  -> shopper: "add it" -> Checkout: add_to_cart, apply_best_coupon(RUN10) -> cart card
  -> Checkout: create_checkout_link -> "Confirm & pay $100.80" (proposal, expires 30 min)
  -> shopper pays on the store's checkout (Stripe / SSLCommerz / bKash) - the agent never holds
card data
  -> payment webhook -> order confirmed -> Order care sends confirmation on WhatsApp
  -> Audit log: intent, picks shown, reasons, coupon, price checks, payment reference
```

05 Guardrails — what stays with people

- **The shopper presses pay.** The agent never stores card details and never completes a payment; it creates a checkout link or summary the shopper confirms.
- **Prices, stock, delivery dates and coupons come from the store system.** Product cards are built from database values; the model's text is checked against them (Nofa's plain-code check pattern).
- **Honest reviews:** summaries quote real reviews and include the negatives; no invented praise; sponsored placements labelled.
- **No dark patterns:** no fake scarcity or countdowns; recommendations explain why.
- **Consent for memory:** sizes and taste are stored only with opt-in; one-tap wipe; data export.
- **Merchant changes are proposals:** listings, prices, offers and purchase orders go live only after the owner approves; margin floors enforced in code.
- **Refund limits:** auto-refund only within policy and under a set amount; above it, staff approve.
- **Messaging rules:** cart-recovery and alerts respect opt-in, quiet hours and frequency caps (WhatsApp template rules).
- **Full audit trail** of what the agent showed, why, and every action taken.

06 Channels

Channel	Use	How (reusing ZOVAK)
Website chat + voice	Shopping, tracking, returns	The Zovak web widget on the site; product, compare and cart cards
WhatsApp	Shopping, order updates, cart recovery	Nofa's WhatsApp Cloud API adapter; catalogue messages
Instagram DM	"Price?" / "Available in M?" comments and DMs	Meta Messenger API (new adapter, same agent)

Channel	Use	How (reusing ZOVAK)
Phone	Order help	Telnyx → OpenAI SIP (Nofa's working setup)
Email	Order confirmations, support inbox	Gmail/Outlook adapter
SMS	Delivery OTP and windows	Alpha SMS for Bangladesh, as planned for Nofa
Merchant console	Store copilot, approvals, insights	Staff-portal pattern from Nofa

07 Data model (first version)

Entity	Key fields
Store (tenant)	id, name, platform (own/Shopify/Woo), policies, margin_rules, refund_limits
Product / Variant	id, title, description, attributes (use, fit, material…), variants[{size, colour, sku, price, stock}], images, embeddings
SizeChart	brand, category, size → measurements
Review	product_id, rating, text, fit_signal, aspects[]
Shopper	id, contacts, consent flags, sizes (opt-in), preferences, addresses
Cart / CartLine	shopper_id, lines[{variant, qty, price_at_add}], coupons, delivery_slot
Order	id (AS-…), lines, totals, payment_ref, status, shipment_ref
Return / Swap / Refund	order_id, lines, reason, policy_check, status, label_ref
Alert	shopper_id, product_id, condition (price ≤ x / back in stock), expires_at
Proposal	kind (payment/refund/swap/listing/price/PO), payload, needs (shopper/staff), status, expires_at
AuditEvent	time, actor, action, inputs, outputs, proposal_id
Conversation / Message	Shared with ZOVAK schema

08 API the website already expects

The website widget is wired for this contract. Turning on NEXT_PUBLIC_AGENT_LIVE=true and pointing the Next.js route at the backend makes the site live without touching the UI.

```
POST /api/agent/chat            (Next.js route -> proxies to the Aisle backend, like Nofa ->
ZOVAK)
request:  { "message": "Running shoes under $120", "conversationId": "web-…",
            "history": [{ "role": "user"|"agent", "text": "…" }] }
response: { "text": "…", "quick": ["What size should I get?", "…"], "cards": [ Card, … ] }

Card =
 | { kind: "products", items: [{ id, name, price, note, icon, tone }] }   // icon/tone -> image
URL later
 | { kind: "compare", a, b, rows: [{ label, a, b }] }
 | { kind: "cart", lines: [{ name, qty, price }], total }
 | { kind: "proposal", title, detail, confirm, done }                     // needs a human tap
```

Backend endpoints (proposed)

Method & path	Purpose
POST /v1/chat (stream)	Conversation turn; streams text, then cards
GET /v1/products/search?q=&filters=	Hybrid search (keywords + embeddings)
POST /v1/cart/{id}/lines · /coupon · /slot	Cart operations
POST /v1/checkout/{cart}/link	Creates the payment link the shopper confirms
POST /v1/proposals/{id}/approve I /reject	Shopper or staff decision; re-validated server-side
GET /v1/orders/{id} · /tracking	Order care
POST /v1/returns	Swap / return / refund request
POST /v1/webhooks/{paymentIcourierImetalshopify}	Signed, de-duplicated inbound events
GET /v1/merchant/insights · /forecast	Store copilot
POST /v1/voice/session	Browser voice session (GPT-Live), same as Nofa

09 Integrations

Need	Options	First choice for MVP
Store platform	Own catalogue, Shopify (Storefront + Admin API), WooCommerce	Own catalogue for the demo, Shopify next
Search	Postgres full-text + pgvector embeddings, Algolia, Meilisearch	Postgres + pgvector
Payments	Stripe (intl), SSLCommerz / bKash / Nagad (Bangladesh), Razorpay (India)	Stripe test mode + SSLCommerz sandbox
Couriers	Pathao, RedX, Steadfast (BD), Shiprocket (IN), EasyPost/Shippo (intl)	One BD courier API

Need	Options	First choice for MVP
Messaging	Meta WhatsApp + Instagram, Telnyx, Gmail, Alpha SMS	Reuse Nofa adapters
Images	Store's product photos; AI try-on later	Store photos
Analytics	Store events + our audit log	Own tables

10 What we reuse from Nofa / ZOVAK

ZOVAK piece	Reused for Aisle as
Channel adapters (web, voice, phone, WhatsApp, SMS, email)	Same adapters; add Instagram DM
Agent loop with tool calling and streaming	Orchestrator + shopping specialists
Read / Propose / Mutate policy, expiring proposals	Payment hand-off, refunds, swaps, listings, POs
Plain-code checks after the model; cards only when relevant	Price/stock/coupon checks before cards render
Outbox worker + notifications	Order updates, alerts, cart recovery with caps
Staff portal	Merchant console with approvals
Audit, idempotency, webhook de-dup	Payments and courier webhooks
Next.js proxy + private demo gate	Same for the Aisle site

> New pieces Aisle needs that Nofa doesn't have: product search with embeddings, size/fit logic, cart & checkout links, payment webhooks, courier tracking, and merchant analytics.

11 Roadmap

Phase	Scope	Done when
0 · Demo site (done)	Website, 3D orb + product ring, scroll story, merchant console mock, chat + voice widget in demo mode	Live on a URL you choose
1 · Shopping agent MVP	Real catalogue (~200 products), search + 3 picks + compare cards, web chat and voice	A shopper finds and carts a product by asking
2 · Checkout + order care	Cart, coupons, payment link (Stripe/SSLCommerz test), courier tracking, returns proposals, WhatsApp	Order placed and tracked end-to-end (test mode)
3 · Size & memory	Size charts, fit signals from reviews, consented profile, alerts	Size recommendation with confidence on every apparel/shoe pick

Phase	Scope	Done when
4 · Store copilot	Listing drafts, review insights, cart recovery, stock forecast, approvals console	Owner approves AI work in one queue
5 · Plug-in mode	Shopify app / WooCommerce plugin with the widget + agent	Installs on a real store

12 How we measure it

Metric	Target for a pilot store
Conversion of chat sessions vs. site average	+20 to +40%
Size-related returns	–20 to –30%
Support tickets handled by the agent	≥ 70%
Abandoned carts recovered	+8 to +12%
Time to publish a product listing	< 1 minute with review
Prices/stock shown that didn't match the store	0

13 Risks

Risk	Mitigation
Agent shows a wrong price or out-of-stock item	Cards built from live data; re-check at add-to-cart and checkout
Pushy or manipulative selling	No fake scarcity; reasons shown; easy "just browsing"
Payment security	Payment only on the provider's hosted checkout; agent never sees card data
Privacy of shopper profiles	Opt-in memory, wipe/export, retention policy
WhatsApp/Instagram policy limits	Templates, opt-in, frequency caps, 24-hour window rules
Integration effort per store	Own catalogue first; then one platform (Shopify) as a plug-in

14 Decisions I need from you

- **Product shape:** our own demo store, or an agent plug-in for existing Shopify/WooCommerce stores (or both, in that order)?

- **Market first:** Bangladesh (SSLCommerz/bKash, Pathao/RedX, Alpha SMS) or international (Stripe, EasyPost)?
- **Catalogue for the pilot:** a real partner store, or a sample catalogue with real product photos you provide?
- **Separate repo or inside ZOVAK?** My suggestion: aisle-backend, importing shared ZOVAK packages, so Nofa stays untouched.
- **Refund auto-limit and cart-recovery caps** to start with.
- **Domain and deploy:** e.g. aisle.zovak.tech on Vercel, like Nofa.